

Empirical Mode Decomposition and Graph Transformation of Financial Series: A Multiscale Topological Analysis for Graph Neural Network Modeling

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Abstract. Financial time series combine long-horizon drift, medium-term cycles, and short-horizon volatility in a single non-stationary path, motivating adaptive multiscale decomposition and relational encodings for downstream graph neural network (GNN) modeling. Existing work, however, treats backend selection among empirical mode decomposition (EMD) variants, oscillatory-panel compression, and time-series-to-graph transformation largely in isolation, relying on heuristic method choice and validation metrics centred on reconstruction or forecast error. We propose an end-to-end pipeline that prepares individual financial instruments for relational analysis under explicit structural budgets. Each series passes an econometric eligibility gate and is decomposed after a two-stage backend selection that couples synthetic pathology benchmarking—covering close-frequency beating, chirps, localized bursts, and equity-like drift—with residual-aware validation on real assets. CEEMDAN is retained for its superior slow-trend purity on a diversified panel comprising MSCI World, sector ETFs (XLE, XLP, XLV), and spot gold. The oscillatory intrinsic mode function panel is then compressed with FastICA to four or five scores with negligible mutual correlation, reducing channel count while preserving multiscale structure. Each retained component is mapped in parallel to natural visibility, horizontal visibility, and recurrence graphs, yielding a structured graph panel for subsequent GNN modeling. Results show that decomposition quality must be judged through trend-component interpretability, not closure error alone, and that ICA leaves the output channels practically independent—a property that supports embarrassingly parallel graph construction and distributed processing of each branch, accelerating pipelines as temporal length and instrument coverage grow.

Keywords: Empirical Mode Decomposition · CEEMDAN · Independent Component Analysis · Visibility Graphs · Recurrence Networks · Graph Neural Networks · Financial Time Series · Multiscale Analysis

1 Introduction

Financial asset prices combine persistent long-horizon drift, medium-term cyclical structure, and short-horizon volatility in a single non-stationary path [10,40]. These stylized facts motivate multiscale analysis, yet classical spectral tools rely on fixed basis functions and are poorly matched to regime shifts, localized bursts, and heavy-tailed fluctuations. Empirical mode decomposition (EMD) and its ensemble extensions address this limitation by adaptively separating oscillatory intrinsic mode functions (IMFs) from a slow residual without prescribing a priori spectral bands [22,50,47]. Variational mode decomposition (VMD) offers a complementary variational route to band-limited separation [15]. In practice, however, backend choice among EMD, EEMD, CEEMDAN, and VMD remains largely heuristic, and comparative evidence is often restricted to reconstruction or forecast error on a narrow set of equity benchmarks [40,31,33,2].

Parallel to decomposition research, time-series-to-graph encodings have emerged as a relational alternative to purely sequential representations. Visibility graphs (NVG and HVG) and recurrence networks map each temporal observation to a node and encode geometric or dynamical proximity through edges, enabling topological characterization of market states and regime transitions [30,36,16,14,43]. Financial studies have applied these encoders primarily to raw prices, returns, or aggregate volatility series [53,45,21]. A smaller body of work maps visibility graphs onto IMF channels extracted by EMD [55,34], yet the literature has not integrated decomposition backend selection, oscillatory-panel compression, and scale-resolved graph encoding into a single reproducible pipeline aimed at downstream graph neural network (GNN) modeling [51,18,5].

Two practical constraints sharpen this gap. First, each retained IMF or ICA score can be encoded independently into multiple graph families; when decomposition yields eight to fourteen modes per asset, the combinatorial cost of graph construction and message passing grows rapidly unless the oscillatory panel is compressed before encoding [7,25,6]. Second, the econometric interpretability of the slow trend component—the primary carrier of long-horizon drift in EMD-family workflows—is seldom treated as a first-class validation target alongside numerical reconstruction [40,2]. Backend comparisons driven only by forecast error can therefore favour spectrally compact decompositions whose trend modes do not align with the monotone residuals required for interpretable multiscale analysis.

This work proposes an end-to-end multiscale pipeline that prepares financial series for relational modeling under explicit structural budgets. Each instrument first passes an econometric eligibility gate that verifies stationarity patterns, non-linear dependence, and time-varying amplitude-frequency structure consistent with effective EMD [40,10]. Eligible series are decomposed with CEEMDAN after a two-stage backend selection that combines synthetic pathology benchmarking with residual-aware validation on real assets. The oscillatory IMF panel is then compressed with FastICA [23,9], reducing channel count while driving cross-correlation among scores toward zero. Each retained independent component is mapped in parallel to NVG, HVG, and recurrence graphs [30,36,14], yielding a

structured graph panel suitable as input to GNN architectures surveyed in the financial forecasting literature [18,5,19]. Figure 1 in Section 3 summarizes the processing chain.

The main contributions are as follows:

1. A synthetic benchmark that subjects EMD, EEMD, CEEMDAN, and VMD to an identical catalog of controlled signal families—close-frequency beating, chirps, localized bursts, noisy superpositions, and a multi-component stress test with equity-like drift—and reports mode-mixing diagnostics jointly with reconstruction fidelity (Section 4.2).
2. Residual-aware empirical validation on a diversified panel spanning MSCI World, three U.S. sector ETFs outside the technology-dominated weighting of global benchmarks (XLE, XLP, XLV), and spot gold (XAUUSD), with trend-component purity assessed through smoothness, monotonicity, and temporal linearity rather than closure error alone (Section 4.3).
3. An explicit ICA compression stage that trades IMF-panel reconstruction fidelity against near-zero cross-correlation among output scores, lowering the number of parallel graph-encoding branches while preserving multiscale oscillatory structure (Section 4.4).
4. A reproducible experimental program that couples the preceding stages with scale-resolved NVG, HVG, and recurrence encodings and structural graph audits, closing the gap between isolated decomposition, compression, and graph-transformation literatures identified in Section 2.4.

The remainder of this paper is organized as follows. Section 2 reviews EMD-family and VMD applications in finance, dimensionality-reduction strategies for multiscale panels, and time-series-to-graph encodings, and synthesizes the research gaps that motivate our experiments. Section 3 formalizes the eligibility gate, CEEMDAN decomposition, ICA compression, and graph-encoding operators. Section 4 reports synthetic and empirical results through the graph-encoding stage. Section 5 concludes and outlines directions for GNN-based forecasting on the generated graph panel.

2 Related Work

2.1 Empirical Mode Decomposition in Financial Time Series

The Multiscale Nature of Financial Markets Financial asset prices typically exhibit pronounced non-stationarity, shifting market regimes, and stochastic drift, effectively behaving as integrated series of order one, or $I(1)$ variables. Conversely, their logarithmic returns generally approximate stationarity but are characterized by high nonlinearity, heavy-tailed distributions, and volatility clustering. These stylized facts are well documented in the empirical finance literature [10,40]. Given the superposition of disparate market forces, from short-horizon volatility fluctuations to medium-term business cycles and long-term macroeconomic trends, financial time series are inherently multiscale.

Classical spectral tools, such as the Fourier transform or discrete wavelet transforms, are bounded by fixed basis functions and uncertainty principles, limiting their efficacy when applied to highly non-stationary and non-Gaussian financial data. This structural limitation necessitates fully data-driven decomposition techniques that adaptively isolate underlying frequency bands without relying on a priori functional bases.

Classic EMD, Ensemble Extensions, and Variational Alternatives To address the rigidities of fixed-basis spectral methods, Empirical Mode Decomposition (EMD) was introduced as a fundamentally heuristic, data-driven sifting process [22]. EMD decomposes any complex signal into a finite sequence of intrinsic mode functions (IMFs) and a monotonic or constant residual. Each IMF must strictly satisfy two conditions: the number of extrema and zero-crossings must be equal or differ by at most one, and the local mean of its upper and lower envelopes must be theoretically zero. In econometric applications, the higher-frequency IMFs represent stochastic market volatility and transient shocks, the lower-frequency IMFs correspond to broader macroeconomic cyclical movements, and the residual encapsulates the underlying secular trend.

Despite its broad applicability, classic EMD suffers significantly from *mode mixing* and boundary/end effects that blur the physical meaning of individual IMFs [41,40]. Mode mixing occurs when a single IMF contains signals of widely disparate scales, or conversely, when a signal of a consistent scale is artificially fragmented across multiple IMFs. Analytical work on two-tone signals shows that EMD separation depends critically on amplitude and frequency ratios [41]; in financial data, spectral leakage is further induced by signal intermittency, localized volatility bursts, and structural breaks.

To mitigate these distortions, Ensemble Empirical Mode Decomposition (EEMD) was developed [50]. EEMD performs the sifting process over an ensemble of trials, each augmented with finite-variance Gaussian white noise, and averages the outcomes so that spurious mode overlaps tend to cancel across realizations. However, EEMD introduces a persistent reconstruction error and may yield a varying number of IMFs across noise realizations, complicating automated modeling pipelines.

These algorithmic deficiencies were addressed by Complete Ensemble Empirical Mode Decomposition with Adaptive Noise (CEEMDAN) [47]. CEEMDAN injects stage-adaptive noise and enforces a complete reconstruction at each sifting stage. It is designed to recover the input signal exactly in the ideal case while yielding more stable IMFs than EEMD. Reported financial applications indicate improved spectral separation relative to classical EMD and residuals that more consistently capture slow drift [40,32]. A well-behaved, approximately monotonic residual remains a critical prerequisite for interpreting long-term deterministic or stochastic financial drift.

Variational Mode Decomposition (VMD) offers a variational alternative that extracts band-limited modes with compact spectral support [15]. Financial studies have applied VMD to exchange-rate volatility modeling [31], multi-step equity

forecasting [39], and volatility-index prediction [27]. Because VMD imposes explicit spectral structure through bandwidth control and prescribed mode count, its extracted DC-like trend modes need not coincide with the monotone residuals produced by EMD-family sifting. Reported method comparisons highlight a tension between compact spectral separation and econometric interpretability of the slow component when backend choice is driven mainly by downstream forecast error [31,39,27,40].

EMD Applications in Financial Analysis and Modeling CEEMDAN has been integrated into multiscale forecasting architectures for global equity markets [33,38,13,32,2]. VMD-based alternatives target index and volatility forecasting [39,27,31], and combined VMD and CEEMDAN pipelines have been applied to cryptocurrency returns [49]. Representative hybrids pair decomposition with LSTM or SVR on equity indices. In EMD-family workflows the residual is typically interpreted as trend, forecast directly, or decomposed further, as in recursive CEEMDAN-STL-SVR models on Nasdaq, Dow, and DAX [2]. High-frequency IMFs are often filtered or routed to volatility-oriented predictors, while lower-frequency cyclical modes and the trend component feed deterministic or recurrent readouts [32,2].

2.2 Dimensionality Reduction and Scalability

After multiscale decomposition (Section 2.1), each retained intrinsic mode function (IMF) can be transformed into a graph for downstream relational analysis (Section 2.3). When the number of components is large, computational cost increases across graph construction and subsequent modeling. Compressing the oscillatory IMF panel from $\mathbb{R}^p$ to $\mathbb{R}^k$ with $k < p$ reduces the number of channels to encode and the aggregate node and edge burden before any relational modeling.

Independent Component Analysis (ICA) is a natural choice for such multichannel panels: under a linear mixing model it seeks statistically independent latent sources whose scores exhibit cross-correlation close to zero, beyond the uncorrelated but not independent outputs of PCA [1,26]. Near-independent channels limit redundant co-movement across parallel encoders and allow each retained score to follow an isolated subgraph pipeline, enabling distributed or embarrassingly parallel processing across workers.

In multivariate finance, ICA has been applied to synchronized panels of asset returns to separate latent shock structures [1]. Chen et al. [7] decompose stock prices with complementary ensemble empirical mode decomposition (CEEMD), restructure the IMFs with sample entropy, and apply ICA before LSTM forecasting on Chinese equities. Jin et al. [25] compress EMD-derived IMFs with PCA prior to neural forecasting on financial indices. Dominant CEEMDAN-LSTM hybrids model each IMF with a separate sequential predictor [6,32].

2.3 Time Series to Graph Transformations

Motivation for Relational Encodings Traditional financial econometrics and deep learning methodologies predominantly treat time series as purely sequential vectors. While sequential models capture temporal autocorrelation and localized patterns, they often struggle to represent the nonlinear geometric structures underlying financial market dynamics. Mapping a one-dimensional time series into a graph topology, where each temporal instance is a node and relational constraints define edges, enables the use of network metrics and topological invariants to characterize market states, regime transitions, and structural breaks [43,56]. This relational perspective can expose higher-order dependencies that are difficult to isolate in standard Euclidean sequence representations.

Visibility Graph Transformations Among deterministic time-series-to-graph methods, visibility algorithms are prominent because they require no free parameters beyond the input series and preserve foundational dynamical properties. Methodological reviews classify univariate mappings into visibility, transition, and proximity families; within visibility methods, the Natural Visibility Graph (NVG) and Horizontal Visibility Graph (HVG) are the canonical constructions [43,30,36].

The NVG connects temporal nodes i and j ($i < j$) when no intermediate point obstructs the line of sight in the time-amplitude plane. This convex mapping relates periodic, random, and chaotic dynamics to characteristic network topologies and preserves fractal scaling, hierarchical structure, and amplitude-ordering information in the original series [30,52]. The HVG simplifies the visibility rule: nodes i and j are adjacent if and only if all intermediate values satisfy $y_k < \min(y_i, y_j)$. HVGs are typically sparser than NVGs and admit exact analytical results for several stochastic processes, which supports discrimination between deterministic and uncorrelated stochastic dynamics [36].

Recent financial applications include multiscale online HVG correlation analysis for scale-dependent cross-market associations [21], rising visibility graphs for recurrent trend motifs in equity indices [54], and visibility-graph degree asymmetries that quantify differences between rising and falling price motions across major international benchmarks [53].

Phase Space Reconstruction and Recurrence Networks A complementary family of encodings derives from dynamical systems theory and recurrence analysis. Eckmann et al. [16] introduced recurrence plots to represent when a reconstructed trajectory revisits similar states in phase space; thresholded recurrence matrices yield graphs whose connectivity summarizes attractor topology [14,56]. Reviews of time-series network science treat recurrence-based mappings alongside visibility and transition methods as standard tools for univariate relational encoding [43,56]. Unlike visibility graphs, which emphasize convex structure in the amplitude domain, recurrence networks foreground dynamical recurrences and are commonly used to probe periodic, quasi-periodic, or deterministic content against uncorrelated noise. Recurrence Quantification Analysis

(RQA) provides widely reported summary measures, including recurrence rate, determinism, and laminarity, that support the study of regime transitions in complex environments [14,56].

Preserved Dynamics and Network Metrics These encoder families emphasize different aspects of the generating process. Visibility graphs tend to reflect fractal and scaling features through heavy-tailed degree distributions and clustering patterns; NVGs are generally more sensitive to amplitude variability and waveform shape than HVGs or recurrence graphs [30,36,55]. Recurrence-based constructions prioritize temporal neighborhood structure in phase space and are often preferred when the goal is to expose determinism, recurrence motifs, or attractor geometry rather than convex visibility in the amplitude domain [14,56].

Empirical finance studies therefore rely on a recurring set of topological summaries: degree distribution and mean degree, edge density and connectivity, average clustering coefficient, characteristic path length, and small-world indicators [52,45,21]. Temporal visibility-graph constructions map successive segments to network states and link consecutive states [45]. Multiplex recurrence networks further enable overlap-based topological summaries linked to early-warning signatures ahead of market crashes [44].

Applications in Financial Markets In financial literature, visibility and recurrence encodings are primarily applied to raw prices, logarithmic returns, or aggregate volatility series on market indices and related benchmarks [53,45]. Representative equity studies cover international indices such as Hang Seng, Dow Jones, FTSE, and SSE Composite [53], as well as U.S. benchmarks in temporally layered visibility constructions [45]. Exchange-rate series map to scale-free or hierarchical visibility networks [52]. Multiscale visibility constructions coarse-grain the series before graph inference [21].

Network-based indicators have been linked to market instability and crisis episodes [42,44]. Recurrence-plot hybrids further support stock trend forecasting [46], and visibility-graph state-transfer models enable evolutionary time-series analysis [45].

Encoding on Decomposed Multiscale Components Most financial encodings operate on aggregate prices or returns [53,45,42]. Zhang [55] applies visibility graphs to IMF series extracted via EMD, and Liu et al. [34] combine IMF-based time-frequency reconstruction with visibility-graph similarity for forecasting. High-frequency IMFs tend to yield different degree, density, and clustering statistics than smooth trends, so graph topology depends on the scale of the input component.

2.4 Synthesis of Research Gaps

The preceding review follows the logical chain of a multiscale pipeline (decomposition, optional panel compression, and graph encoding), yet the literature has

developed these stages largely in isolation. Backend selection, empirical validation, and compression of the oscillatory IMF panel are rarely treated as coupled design choices. Prior work predominantly ranks EMD-family and variational backends from downstream forecast error rather than from unified intrinsic diagnostics, and seldom stress-tests competing methods under controlled synthetic regimes before applying them to financial data [40,31,27]. The gaps below motivate the experimental program developed in this work.

Synthetic Benchmarking of Decomposition Backends Although EMD, EEMD, CEEMDAN, and VMD are widely adopted in financial econometrics, comparative selection among them remains largely heuristic. Analytical studies on two-tone superpositions show that mode separation depends critically on amplitude and frequency ratios [41], but financial applications rarely reproduce this controlled setting at scale. Existing comparisons seldom span multiple synthetic families that emulate market pathologies, including close-frequency beating, linear chirps, localized bursts on carriers, additive noise, and equity-like drift. What is missing is an experimental block that subjects the most popular decomposition backends to the same catalog of synthetic regimes and reports intrinsic quality jointly with reconstruction fidelity. This work supplies such a benchmark: every backend is evaluated on identical signal families so that backend choice for financial series rests on observed behavior under known pathologies rather than on isolated index-level forecasts.

Empirical Validation Beyond Reconstruction Error After synthetic screening, practical validation on real assets remains biased toward broad equity indices and exchange-rate series [40,31,33,2]. Sector exchange-traded funds outside the technology weighting that dominates global benchmarks, and spot gold as a partially decorrelated safe-haven asset, appear less frequently as primary decomposition case studies. Moreover, empirical studies typically privilege aggregate reconstruction error or downstream prediction metrics, while the econometric interpretability of the slow component receives less systematic attention. The monotonic residual is the primary carrier of long-horizon drift in EMD-family workflows [40,2], yet its shape, smoothness, and stability across backends are seldom inspected as first-class validation targets. This work extends the practical experiment to a heterogeneous panel spanning a global market index, sector ETFs from non-dominant industries, and spot gold, and evaluates decomposition quality through direct observation of the generated residual, complementing reconstruction error with residual-focused diagnostics that foreground trend purity.

ICA Compression for Scalable Downstream Processing Isolated hybrids compress EMD-derived panels with PCA or ICA before neural forecasting [25,7], yet dominant CEEMDAN-LSTM pipelines model each IMF with a separate sequential predictor without reducing panel dimension [6,32]. When each retained

component is later encoded as a graph and processed in parallel, the number of channels directly governs aggregate computational cost. No established framework quantifies how ICA-based reduction of the oscillatory IMF panel trades reconstruction fidelity against near-zero cross-correlation among output scores and against the feasibility of embarrassingly parallel, distributed processing per component. This work introduces an explicit ICA compression stage that lowers channel count before downstream relational encoding, minimizes redundant co-movement across parallel workers, and thereby reduces the resource burden of later pipeline phases without abandoning the multiscale structure extracted at decomposition.

Integrated Contribution Taken together, these elements address a gap that prior literature has only partially covered: decomposition backends are chosen heuristically, validated mainly through reconstruction or forecast error on narrow asset sets, and passed forward at full IMF cardinality. The present work closes this gap by coupling a synthetic pathology benchmark, residual-aware empirical validation on diversified financial series, and ICA compression of the oscillatory panel into a single, reproducible experimental program that prepares scale-resolved graph encoding under explicit structural budgets.

3 Proposed Methodology

3.1 Overview

The proposed workflow decomposes financial series into multiscale components, compresses the oscillatory IMF panel, and encodes each retained channel as a graph for downstream graph neural network (GNN) modeling. Figure 1 summarizes the end-to-end processing chain.

Raw financial time series first undergo an eligibility check that verifies whether each series satisfies the conditions required for stable empirical mode decomposition. Series that pass this gate are decomposed into intrinsic mode functions (IMFs) and a slow residual. The oscillatory IMF panel is then compressed with independent component analysis (ICA), which reduces the number of output channels while driving cross-correlation among components toward zero. Each retained ICA score is processed in parallel along an isolated branch and mapped to three complementary graph representations (NVG, HVG, and recurrence) through the encoders defined in Section 3.4. The resulting graph panel constitutes the structured input prepared for downstream GNN modeling.

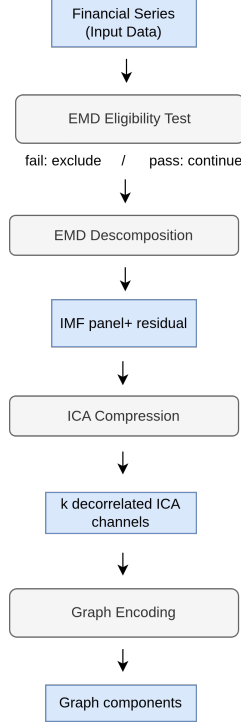


Fig. 1. Proposed multiscale pipeline. Blue boxes: data artifacts; gray boxes: processing operations.

3.2 Multiscale Decomposition with CEEMDAN

Let $x(t)$ denote a univariate financial series observed at $t = 1, \dots, T$ (typically log-prices or detrended log-returns after the eligibility gate in Section 3.1). The objective of the decomposition stage is to separate oscillatory structure from slow drift without imposing a fixed spectral basis [22,40]. Following the synthetic benchmark described in Section 2.4, CEEMDAN is adopted as the operational backend because it mitigates mode mixing of classical EMD and enforces near-exact reconstruction relative to EEMD [50,47].

Empirical mode decomposition. EMD expresses $x(t)$ as a finite sum of intrinsic mode functions (IMFs) plus a residual [22]:

$$x(t) = \sum_{i=1}^n c_i(t) + r_n(t), \quad (1)$$

where $c_i(t)$ are zero-mean oscillatory components ordered from high to low local frequency, and $r_n(t)$ is monotone or nearly constant. Each IMF must satisfy (i)

the number of extrema and zero-crossings differs by at most one, and (ii) the mean of the upper and lower spline envelopes is approximately zero at every t . Extraction proceeds by *sifting*: starting from $h_0(t) = x(t)$, envelopes are interpolated from local maxima and minima, the local mean $m(h)$ is subtracted, and the procedure iterates on the residue until an IMF stop criterion is met [22,41]. Symbolically,

$$h_{j+1}(t) = h_j(t) - m(h_j(t)), \quad (2)$$

and the i -th IMF is the converged h_j after sifting; the next component is obtained by replacing x with $x - \sum_{k < i} c_k$.

Complete ensemble EMD with adaptive noise. CEEMDAN refines EEMD by adding realizations of adaptive noise at each sifting stage and averaging across an ensemble of I trials [47]. Let $E_1(\cdot)$ denote the operator that returns the first IMF of its argument by sifting, and let $n^{(j)}(t)$, $j = 1, \dots, I$, be independent standard Gaussian draws. With noise scale ε and $\sigma(\cdot)$ the sample standard deviation, the first mode is

$$c_1(t) = \frac{1}{I} \sum_{j=1}^I E_1 \left(x(t) + \varepsilon \sigma(x) n^{(j)}(t) \right), \quad (3)$$

and the first-stage residual is $r_1(t) = x(t) - c_1(t)$. For $k \geq 2$, CEEMDAN forms

$$c_k(t) = \frac{1}{I} \sum_{j=1}^I E_k \left(r_{k-1}(t) + \varepsilon \sigma(r_{k-1}) E_k(n^{(j)}(t)) \right), \quad (4)$$

where $E_k(\cdot)$ extracts the k -th IMF of its argument, and updates $r_k(t) = r_{k-1}(t) - c_k(t)$ until a stopping rule is reached (maximum IMF count or negligible oscillatory content in r_k). By construction, $\sum_{i=1}^n c_i(t) + r_n(t)$ reconstructs $x(t)$ up to numerical tolerance, while stage-dependent noise reduces spurious mode alignment across scales [47,40]. In the pipeline, the oscillatory panel $\{c_1, \dots, c_{n-1}\}$ feeds the ICA compression stage (Section 3.3), whereas r_n is retained separately as the slow trend carrier for residue-aware validation.

3.3 IMF-Panel Compression with ICA

Let $\mathbf{X} \in \mathbb{R}^{T \times p}$ denote the matrix of oscillatory IMFs (one column per mode, rows aligned in time), excluding the CEEMDAN residual. ICA compresses $\mathbf{X}$ to $\mathbf{Z} \in \mathbb{R}^{T \times k}$ with $k < p$ under the instantaneous linear mixing model $\mathbf{x}_t = A \mathbf{s}_t$, where $\mathbf{x}_t$ is the t -th row of $\mathbf{X}$, $A \in \mathbb{R}^{p \times p}$ is a full-rank mixing matrix, and $\mathbf{s}_t$ contains statistically independent sources [9,24]. Unlike PCA, which yields orthogonal but not necessarily independent scores [26], ICA seeks a demixing transform W such that $\mathbf{z}_t = W \mathbf{x}_t$ has pairwise correlations close to zero and maximal statistical independence across columns [1,7].

We apply FastICA [23], which estimates W by maximizing a contrast function of non-Gaussianity after centering and whitening of $\mathbf{X}$. Retaining the top

k components yields a decorrelated panel $\mathbf{Z}$ that approximates the original subspace when $k < p$ and reconstructs the standardized IMF block via the fitted unmixing operator. The residual r_n is kept in a separate column and is not mixed into $\mathbf{Z}$. Component count k is selected to balance reconstruction fidelity against low mean absolute cross-correlation among ICA scores; the resulting channels feed independent parallel graph encoders in Section 3.4.

3.4 Time-Series-to-Graph Encoding

Each retained ICA score defines a univariate channel $\{y_t\}_{t=1}^T$ (the residue r_n may be encoded separately for validation). Every channel is mapped to a graph $G = (V, E)$ with $V = \{1, \dots, T\}$ and scalar node feature y_t . Three complementary encoders are applied in parallel to each channel, following the visibility and recurrence families reviewed in Section 2.3 but operating here on decomposed scales rather than on raw prices [55,30,36,14].

Visibility graph encodings. Visibility graphs require no free parameters beyond the input series [30,36]. For nodes $i < j$, the natural visibility graph (NVG) sets

$$A_{ij}^{\text{NVG}} = \mathbb{K}\left[y_k < y_i + \frac{t_k - t_i}{t_j - t_i}(y_j - y_i) \ \forall k \in \{i+1, \dots, j-1\}\right], \quad (5)$$

so that no intermediate sample blocks the line of sight in the time-amplitude plane. The horizontal visibility graph (HVG) applies the stricter rule

$$A_{ij}^{\text{HVG}} = \mathbb{K}[y_k < \min(y_i, y_j) \ \forall k \in \{i+1, \dots, j-1\}], \quad (6)$$

which yields sparser graphs that preserve convex amplitude structure while facilitating analytical comparison across stochastic and deterministic regimes [36]. Directed visibility edges are symmetrized into an undirected edge set for downstream analysis.

Recurrence graph encoding. Recurrence networks summarize dynamical recurrences in reconstructed phase space [16,14]. Given delay τ and embedding dimension d , each time index i is embedded as

$$\mathbf{u}_i = [y_i, y_{i+\tau}, \dots, y_{i+(d-1)\tau}]^\top, \quad (7)$$

and nodes i and j are adjacent when their embedded states are ε_{rec} -close:

$$A_{ij}^{\text{REC}} = \mathbb{K}[\|\mathbf{u}_i - \mathbf{u}_j\|_2 \leq \varepsilon_{\text{rec}}], \quad (8)$$

with the Euclidean norm $\|\cdot\|_2$. Parameters are estimated *per channel*: τ is taken at the first local minimum of mutual information between y_t and $y_{t+\tau}$; d is selected by false-nearest-neighbor (FNN) ratios on the delay coordinates; and ε_{rec} is set to the 10th percentile of pairwise embedding distances when no absolute threshold is supplied. The symmetrized recurrence matrix (without self-loops) defines the final edge set.

Graph panel and structural summaries. For k ICA scores and optional residue encoding, the encoder stage produces up to $3k$ visibility and recurrence graphs on oscillatory channels (plus three graphs on r_n when validated). Each instance is represented by node features $\mathbf{x} \in \mathbb{R}^{T \times 1}$ and a sparse edge index of shape $2 \times |E|$, suitable as input to graph neural networks. Exploratory structural audits record $|V|$, $|E|$, edge density, the number of connected components, mean degree, and, when tractable, average clustering, enabling scale-wise comparison of graph sparsity before GNN modeling (Section 4.5).

4 Experiments and Results

4.1 Data and Experimental Setup

Empirical experiments are conducted on five daily financial series that share a common calendar and are processed *independently*: each asset passes through the eligibility gate, CEEMDAN decomposition, ICA compression, and graph encoding without joint modeling across instruments. This design avoids multivariate coupling at the decomposition and encoding stages, so observed differences reflect asset-specific dynamics rather than cross-sectional mixing.

Empirical panel and sampling window The empirical panel comprises five daily closing-price series: the MSCI World global equity index, three U.S. sector exchange-traded funds (XLE, energy; XLP, consumer staples; XLV, health care), and spot gold (XAU/USD). All instruments share the same sampling window from January 12, 2012 to April 20, 2026 ($T = 3587$ trading days), covering multiple market regimes including sustained bull phases, corrections, and high-volatility episodes. Daily percentage returns are computed for diagnostic testing; the decomposition and graph stages operate on the aligned price levels (or log-transformed variants after the eligibility gate in Section 3.1), following the financial EMD practice surveyed in Section 2.1.

The asset set was chosen to obtain sufficient pattern diversity for general conclusions on financial time series, in line with the diversified validation program motivated in Section 2.4. MSCI World captures the dominant global equity trend, strongly shaped by U.S. large-cap technology weights. XLE probes cyclical, commodity-linked sector dynamics; XLP represents defensive consumption with comparatively smooth demand exposure; and XLV reflects health-care cycles driven by regulation and innovation. Spot gold adds a safe-haven commodity partially decorrelated from equity benchmarks. Together, these instruments stress the pipeline under trend-dominated, sector-rotational, defensive, and commodity-linked behaviors without relying on a single benchmark narrative.

EMD eligibility diagnostics Before decomposition, each series is evaluated with the same three-block testing battery applied independently per instrument. The blocks mirror the prerequisites for effective EMD on financial data [40,10]

and the eligibility gate introduced in Section 3.1. Unless stated otherwise, hypothesis tests are evaluated at significance level $\alpha = 0.05$. The three blocks are defined as follows.

Stationarity. Closing prices and daily percentage returns are tested with the Augmented Dickey-Fuller (ADF) test [12] (automatic lag selection by AIC) and the Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test [29] with trend-stationary null and automatic bandwidth. EMD is intended for level-non-stationary series with stationary increments. On *closing prices*, the requirement is met when ADF fails to reject the unit-root null ($p > \alpha$) and KPSS rejects trend stationarity ($p < \alpha$). On *daily percentage returns*, the complementary pattern is required: ADF rejects a unit root ($p < \alpha$) while KPSS does not reject stationarity ($p > \alpha$). Joint satisfaction of both tests on prices and returns indicates a persistent trend in levels with mean-reverting short-horizon fluctuations.

Nonlinearity. Nonlinear structure is assessed via the BDS test [4] on closing prices, applied to AR(1) residuals across embedding dimensions $m = 2, \dots, 5$. The nonlinearity requirement is satisfied when BDS rejects the independence null at least at dimensions $m = 3, 4, 5$ ($p < \alpha$), indicating structure beyond linear autocorrelation. Complementary evidence comes from squared-return dependence: the Ljung-Box test on r_t^2 [35] should reject the absence of autocorrelation ($p < \alpha$), signaling volatility clustering consistent with ARCH-type dynamics [17,3]. A GARCH(1,1) fit with lower AIC/BIC than ARCH(1) on percentage returns provides additional support but is not treated as a hard exclusion criterion.

Time-varying amplitude and frequency. Because EMD is designed for oscillations with evolving scale, each series is further characterized by rolling 252-day volatility (one trading year) and by zero-crossing counts in moving windows of the same length. Amplitude variability is summarized by the coefficient of variation (CV) of the rolling volatility series; values above 0.3 are interpreted as high time-varying amplitude. Frequency variability is summarized by the mean number of zero-crossings per window together with the CV of window-wise crossing rates, where values above 0.3 indicate materially evolving local frequency. Elevated values on both dimensions support the presence of multiple active temporal scales in the observed path.

Empirical outcomes. Table 1 reports the numerical outcomes of the eligibility battery for each asset ($T = 3587$). Test statistics are shown with p -values in parentheses where applicable. Across the full panel, every instrument satisfies the stationarity pattern on prices and returns, exhibits significant BDS and Ljung-Box evidence at $\alpha = 0.05$, and displays amplitude CV values above 0.3 together with sustained zero-crossing activity. All five series pass the eligibility gate and enter the decomposition stage.

In summary, persistent trends in prices, stationary returns, nonlinear dependence, volatility clustering, and evolving amplitude-frequency structure are confirmed numerically for every instrument. The empirical program then proceeds

Table 1. EMD eligibility diagnostics by asset (independent testing per series, $T = 3587$). BDS is applied to AR(1) residuals on closing prices. Amplitude variability is the coefficient of variation of 252-day rolling volatility; frequency variability is the mean zero-crossing count per moving window.

Test	MSCI	XLE	XLP	XLV	XAU
Stationarity (closing prices)					
ADF	1.66 ($p = 0.998$)	-0.08 ($p = 0.952$)	-0.55 ($p = 0.882$)	-0.61 ($p = 0.870$)	2.96 ($p = 1.000$)
KPSS	1.33 ($p = 0.010$)	1.61 ($p = 0.010$)	0.87 ($p = 0.010$)	0.70 ($p = 0.010$)	1.40 ($p = 0.010$)
Stationarity (daily returns, %)					
ADF	-19.73 ($p = < 0.001$)	-11.68 ($p = < 0.001$)	-18.89 ($p = < 0.001$)	-16.47 ($p = < 0.001$)	-62.65 ($p = < 0.001$)
KPSS	0.03 ($p = 0.100$)	0.05 ($p = 0.100$)	0.02 ($p = 0.100$)	0.03 ($p = 0.100$)	0.05 ($p = 0.100$)
Nonlinearity (BDS, AR(1) residuals)					
$m = 3$	21.27 ($p = < 0.001$)	17.28 ($p = < 0.001$)	22.71 ($p = < 0.001$)	21.53 ($p = < 0.001$)	24.85 ($p = < 0.001$)
$m = 4$	25.28 ($p = < 0.001$)	20.75 ($p = < 0.001$)	25.54 ($p = < 0.001$)	24.93 ($p = < 0.001$)	26.37 ($p = < 0.001$)
$m = 5$	28.35 ($p = < 0.001$)	23.42 ($p = < 0.001$)	27.84 ($p = < 0.001$)	27.72 ($p = < 0.001$)	27.44 ($p = < 0.001$)
Volatility clustering (Ljung-Box, r_t^2)					
Ljung-Box	3236.82 ($p = < 0.001$)	2267.29 ($p = < 0.001$)	5072.60 ($p = < 0.001$)	4816.87 ($p = < 0.001$)	318.03 ($p = < 0.001$)
Time-varying amplitude and frequency					
Vol. CV (252d)	0.597	0.462	0.378	0.427	0.952
Mean crossings	12.04	12.67	13.95	13.31	11.04

asset by asset on a homogeneous calendar with a shared methodological backend, while preserving cross-asset diversity in the underlying financial patterns.

4.2 Synthetic Benchmark and Decomposition Model Selection

Following the synthetic pathology program outlined in Section 2.4, five controlled signal families are generated ($T = 2500$, sampling rate 500 Hz, duration 5 s): close-frequency tones (amplitude beating), a linear chirp, a high-frequency burst on a low-frequency carrier, a noisy superposition of close tones and chirp, and a multi-component stress test that aggregates mode-mixing, ascending and descending chirps, close tones, two isolated sinusoids, Gaussian noise, and a slow linear ramp mimicking equity-like drift [41]. Figure 2 illustrates the resulting waveforms. Each family is decomposed with EMD, EEMD, CEEMDAN, and variational mode decomposition (VMD) [15,50,47]. Ensemble hyperparameters for EEMD, CEEMDAN, and VMD are calibrated on the multi-component stress test to limit cross-mode linear coupling while preserving numerical reconstruction; EMD serves as a non-ensemble reference with a shared cap of twelve modes.

Mode-mixing diagnostics. Let r_{ij} denote the sample Pearson correlation between oscillatory modes i and j (slow trend component excluded), computed over the first $\min\{8, N_{\text{IMF}}\}$ modes. For each IMF pair we test $H_0: \rho_{ij} = 0$ at $\alpha = 0.05$ without multiple-comparison adjustment. Mode mixing is summarized by (i) the *median* of $|r_{ij}|$ across pairs, which is robust to a few highly coupled pairs, and (ii) $\pi_{0.05}$, the percentage of pairs with $p < 0.05$. Lower values on both metrics indicate weaker linear coupling and therefore less mode mixing. All methods achieve negligible relative reconstruction error ($\|\hat{x} - x\|/\|x\| < 10^{-15}$ in every run), so differences reflect scale separation rather than poor closure. Table 2 reports the resulting statistics for each signal family.

Table 2. Synthetic benchmark: mode mixing by method and signal family. Each cell reports median $|r_{ij}|$ across IMF pairs; the parenthetical value is $\pi_{0.05}$ (% of pairs with $p < 0.05$). Lower is better.

Family	EMD	EEMD	CEEMDAN	VMD
Close tones	0.038 (40%)	0.011 (21%)	0.050 (57%)	0.002 (13%)
Linear chirp	0.009 (33%)	0.010 (29%)	0.018 (47%)	0.038 (47%)
Burst on carrier	0.057 (67%)	0.009 (29%)	0.006 (29%)	0.070 (73%)
Close tones + chirp + noise	0.028 (43%)	0.008 (29%)	0.012 (19%)	0.004 (27%)
Multi-component stress	0.010 (19%)	0.010 (32%)	0.007 (24%)	0.004 (7%)

Benchmark outcomes. On the drift-rich multi-component stress test, VMD exhibits the weakest coupling (median 0.004, $\pi_{0.05} = 7\%$), followed by CEEMDAN (median 0.007, $\pi_{0.05} = 24\%$), EMD (median 0.010, $\pi_{0.05} = 19\%$), and EEMD (median 0.010, $\pi_{0.05} = 32\%$). VMD also ranks first on the noisy superposition

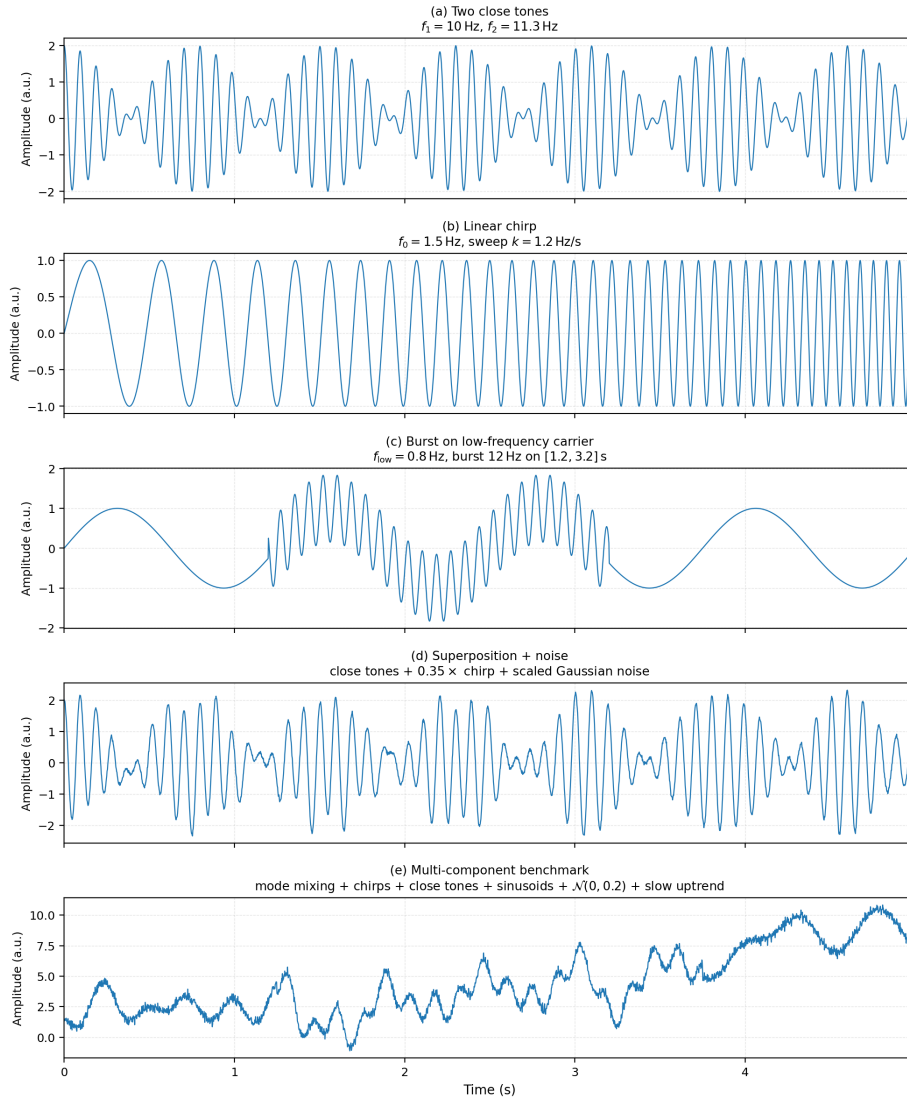


Fig. 2. Synthetic signals used in the decomposition benchmark ($T = 2500$, 500 Hz, 5 s). **(a)** Close tones at 10 Hz and 11.3 Hz (amplitude beating). **(b)** Linear chirp with $f_0 = 1.5 \text{ Hz}$ and sweep rate 1.2 Hz/s . **(c)** 0.8 Hz carrier with a 12 Hz burst on $t \in [1.2, 3.2] \text{ s}$. **(d)** Superposition of close tones, $0.35 \times$ chirp, and scaled Gaussian noise. **(e)** Multi-component stress test: mode-mixing burst, ascending and descending chirps, close tones, isolated sinusoids, $\mathcal{N}(0, 0.2)$ noise, and slow linear uptrend.

(median 0.004, $\pi_{0.05} = 27\%$). No method dominates every family: on close-frequency tones, for example, VMD shows a low median but a comparatively

high mean absolute correlation, whereas CEEMDAN records the largest $\pi_{0.05}$ on that family. Averaged across all five families, EEMD attains the lowest mean median $|r_{ij}|$ (0.009), while VMD and CEEMDAN trade the lead on the stress scenarios most relevant to financial indices.

Backend selection. Taken together, the synthetic evidence shows that VMD can deliver the sparsest cross-mode correlation structure on the aggregated stress test and other challenging superpositions, which is consistent with its variational band-separation objective [15]. Nevertheless, CEEMDAN is adopted as the operational backend for the empirical assets in Section 4.3. When all four methods are applied to the financial panel, CEEMDAN yields slow trend components that concentrate long-horizon drift and support trend-purity validation, whereas VMD tends to absorb trend energy into a narrow DC mode with weak oscillatory closure on the same synthetic stress family (slow-trend energy fraction $\approx 0.4\%$ for VMD versus $\approx 75.5\%$ for CEEMDAN). The comparative behavior of slow trend components on real prices is analyzed in Section 4.3. This two-stage selection, synthetic coupling first and empirical trend-component purity second, aligns the decomposition choice with both mode-separation diagnostics and the interpretability requirements of the multiscale pipeline.

4.3 Decomposition Outcomes on Empirical Assets

Each eligible asset is decomposed independently with CEEMDAN, EEMD, and VMD. The subsections below report hyperparameter calibration, validation of the decompositions, and a synthesis of the empirical evidence.

Hyperparameter calibration Hyperparameter selection follows a two-stage logic that mirrors the backend choice in Section 4.2: synthetic signals first establish ensemble settings that limit mode mixing, and real prices then refine asset-specific rules where trend interpretability matters. Accordingly, calibration is *method-specific* on the empirical panel: CEEMDAN and VMD are tuned independently for each instrument, whereas EEMD is held at one shared configuration and treated as a fixed-ensemble reference.

For *CEEMDAN*, the search spans $\varepsilon \in [0.03, 0.12]$ for the noise standard deviation, $N_{\text{trials}} \in [100, 220]$ for the ensemble size, and $\text{max_imf} \in [9, 14]$ for the mode cap; MSCI World additionally retains the exploratory reference tuple ($\varepsilon = 0.05$, $N_{\text{trials}} = 100$, $\text{max_imf} = 14$). Configurations failing the reconstruction gate ($\text{RMSE}_{\text{rel}} \geq 10^{-10}$) are discarded. Among the survivors, the optimum is the setting whose explicit slow trend component is most trend-like, prioritizing upward monotonicity, temporal linearity (R^2 of a least-squares line), and the fraction of positive first differences; remaining ties favor lower linear-fit error and weaker linear coupling among IMFs.

By contrast, *EEMD* is not re-optimized on each asset. All empirical series share a single PyEMD configuration ($\text{noise_width} = 0.05$, $N_{\text{trials}} = 100$, $\text{SD_thresh} = 0.25$, $\text{max_imf} = 14$) aligned with the MSCI reference implementation, which isolates differences in trend-component purity from differences

in ensemble tuning. On the synthetic stress test (Section 4.2), a separate tuple ($\text{noise_width} = 0.095$, $\text{SD_thresh} = 0.18$) was calibrated to minimize mode mixing on the multi-component benchmark.

With respect to *VMD*, it is scanned over $K \in [6, 12]$ modes and $\alpha \in [10^3, 5 \times 10^3]$ for the bandwidth penalty. The *DC* flag is fixed to 1 throughout: when $\text{DC} = 1$, VMD extracts a dedicated zero-frequency component (the direct-current, or mean-trend, mode) as its first mode [15]. This design is deliberate for cross-method comparison, because CEEMDAN and EEMD isolate long-horizon drift in an explicit or implicit slow trend component, whereas VMD with $\text{DC} = 0$ would spread trend energy across band-limited modes without a comparable slow trend component. With $\text{DC} = 1$, IMF_1 acts as the slow trend component and $\sum_{i \geq 2} \text{IMF}_i$ as the oscillatory part, mirroring the CEEMDAN/EEMD partition shown in Figure 3. The selected (K, α) pair minimizes mean absolute correlation between modes, with ties resolved by lower slow-trend energy fraction; every retained solution closes the signal numerically ($\text{RMSE}_{\text{rel}} \approx 0$ in double precision).

Validation of decompositions Decompositions are validated jointly on numerical closure, trend-component purity, and visual scale allocation. Reconstruction fidelity is assessed by $\text{RMSE}_{\text{rel}} = \|\hat{x} - x\|_2 / \|x\|_2$. Trend quality is evaluated on the slow trend component associated with each method: the explicit CEEMDAN slow trend component; the implicit EEMD slow trend component $|x - \sum_i \text{IMF}_i|$; and the VMD slow trend component IMF_1 with $\text{DC} = 1$. For each component we report (i) R_{lin}^2 , the coefficient of determination of a linear fit in time; (ii) *roughness* $\mathcal{R}_{\Delta^2}$, the standard deviation of the normalized second difference (values in scientific notation when $\ll 1$); (iii) fraction of positive first differences (Frac. incr.), as a soft monotonicity score; and (iv) Spearman correlation with the time index (ρ_S). Figure 3 complements these metrics with a panel-wise comparison of oscillatory IMF sums (top row) and slow-trend gaps (bottom row) across the five eligible assets.

Table 3 summarizes the quantitative validation by asset and method.

Across methods and assets, RMSE_{rel} in Table 3 is at or below 10^{-6} , so differences among backends reflect *scale allocation*, not failure to close the signal.

Interpretation Table 3 and Figure 3 converge on a single question: how cleanly each method isolates long-horizon drift from oscillatory volatility. On every asset, CEEMDAN delivers the purest slow trend component— R_{lin}^2 between 0.49 and 0.99 and $\mathcal{R}_{\Delta^2}$ between 10^{-7} and 10^{-6} (two to five orders of magnitude below EEMD and VMD)—producing a component that is approximately linear, visually smooth, and genuinely slow-frequency on the strongly drift-dominated series. On XLE and XAU/USD, where drift is less linear and ρ_S is not uniformly highest, CEEMDAN still attains the lowest roughness among the three backends.

EEMD and VMD illustrate contrasting failure modes. The EEMD slow trend component rises and falls in near-equal proportion (Frac. incr. ≈ 0.51), confirming high-frequency leakage into the lower panels of Figure 3. VMD assigns drift to IMF_1 yet retains visible oscillatory structure in the slow-trend gap, notably

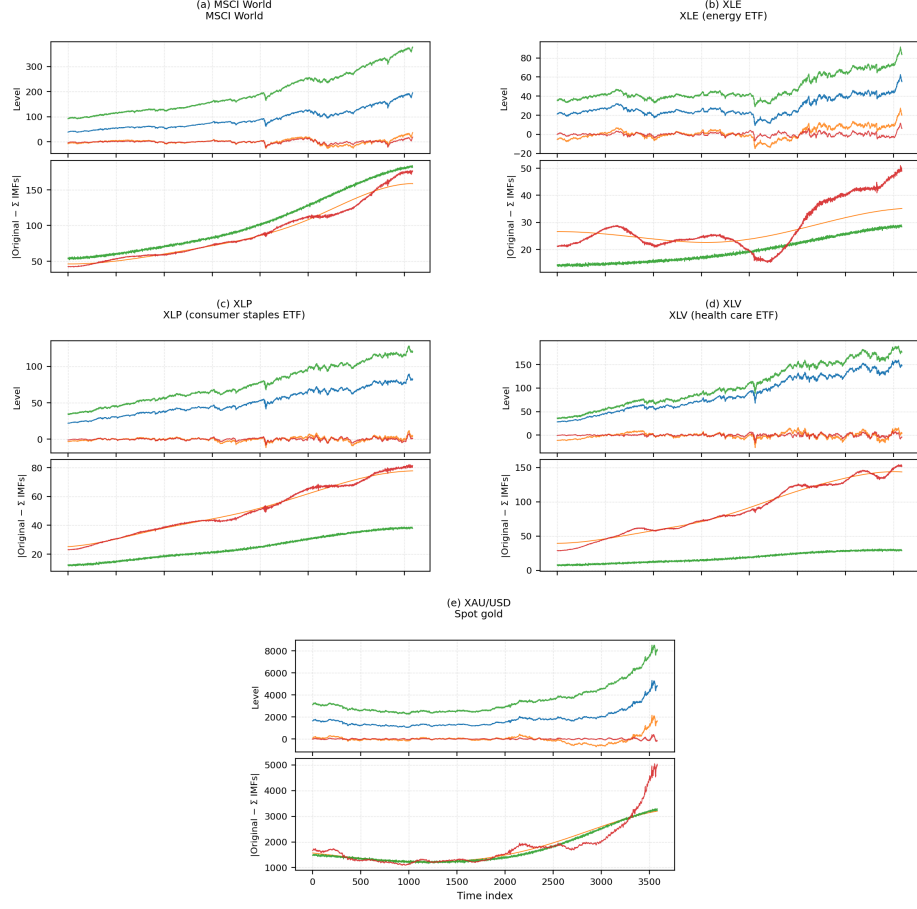


Fig. 3. Comparative decomposition on the empirical panel. Each panel shows two aligned rows: *Level* (top) and $|Original - \sum IMFs|$ (bottom). **Blue:** original closing prices (top row only). **Orange:** CEEMDAN; top row, sum of oscillatory IMFs (slow trend component excluded); bottom row, explicit slow trend component $|x - \sum_i IMF_i|$. **Green:** EEMD; top row, sum of oscillatory IMFs; bottom row, implicit slow trend component $|x - \sum_i IMF_i|$. **Red:** VMD with $DC = 1$; top row, $\sum_{i \geq 2} IMF_i$; bottom row, $|x - \sum_{i \geq 2} IMF_i|$ (VMD slow-trend gap).

on XLE and XAU/USD. CEEMDAN avoids both defects: long-horizon drift is confined to an explicit slow trend component, while the summed oscillatory IMFs—centered near zero in the upper panels—capture essentially all short- and medium-horizon volatility characteristic of these financial series. That partition between a smooth trend and complete oscillatory closure is the behavior required for interpretable multiscale graph encoding. CEEMDAN is therefore retained as the operational backend for the remainder of the pipeline.

Table 3. Trend-component purity by asset and method. Roughness values are reported in scientific notation when small. Despite comparable reconstruction errors, CEEMDAN consistently yields the purest slow trend component (lowest $\mathcal{R}_{\Delta^2}$), indicating cleaner separation of multi-scale oscillations.

Asset	Method	R_{lin}^2	$\mathcal{R}_{\Delta^2}$	Frac. incr.	ρ_S	RMSE _{rel}
<i>MSCI World</i>						
	CEEMDAN	0.949	1.28×10^{-6}	0.992	1.000	6.8×10^{-7}
	EEMD	0.955	4.83×10^{-2}	0.516	1.000	6.8×10^{-7}
	VMD	0.930	1.11×10^{-4}	0.932	1.000	0
<i>XLE</i>						
	CEEMDAN	0.489	1.96×10^{-6}	0.568	0.547	1.3×10^{-18}
	EEMD	0.943	1.40×10^{-1}	0.513	0.995	0
	VMD	0.465	2.10×10^{-4}	0.690	0.531	0
<i>XLP</i>						
	CEEMDAN	0.993	9.11×10^{-7}	1.000	1.000	1.5×10^{-18}
	EEMD	0.988	1.01×10^{-1}	0.514	0.999	0
	VMD	0.984	1.34×10^{-4}	0.918	0.999	0
<i>XLV</i>						
	CEEMDAN	0.982	1.04×10^{-6}	0.977	1.000	1.4×10^{-18}
	EEMD	0.974	2.13×10^{-1}	0.510	0.993	0
	VMD	0.976	9.42×10^{-5}	0.821	0.994	0
<i>XAU/USD</i>						
	CEEMDAN	0.709	1.16×10^{-6}	0.666	0.753	1.4×10^{-18}
	EEMD	0.670	8.09×10^{-2}	0.505	0.720	0
	VMD	0.502	2.73×10^{-4}	0.604	0.760	0

4.4 Dimensionality Reduction of the IMF Panel

After CEEMDAN, each asset yields $p \in \{8, 9\}$ oscillatory IMFs plus an explicit slow trend component. FastICA compresses the oscillatory block to k statistically independent scores $Z_1, \dots, Z_k$. The slow trend component is excluded from the fit and retained separately. The same procedure is applied asset by asset.

Hyperparameter calibration For each asset, k is chosen from $\{2, \dots, p - 1\}$. Inputs are standardized column-wise. FastICA uses the parallel algorithm, unit-variance whitening, and `random_state = 42`. A candidate k is *viable* when $\max_{i \neq j} |r(Z_i, Z_j)| < 0.01$. Among viable values, the chosen k is the *smallest* that simultaneously (i) achieves mean IMF reconstruction R^2 (column-wise) at least 72% of the best viable R^2 , and (ii) keeps the relative Frobenius error within 115% of the elbow reference at the R^2 knee. If no candidate satisfies both constraints, the knee value is retained. On XLP, where every grid point already meets the decorrelation threshold, $k = 4$ is fixed to match the MSCI World reference run; MSCI World itself is evaluated at the same $k = 4$.

Validation and outcomes Figure 4 arranges the ICA scores in a 5×5 matrix (rows: $Z_1, \dots, Z_5$; columns: one per series). Selected dimensionalities are $k = 4$ for MSCI World and XLP and $k = 5$ for XLE, XLV, and XAU/USD.

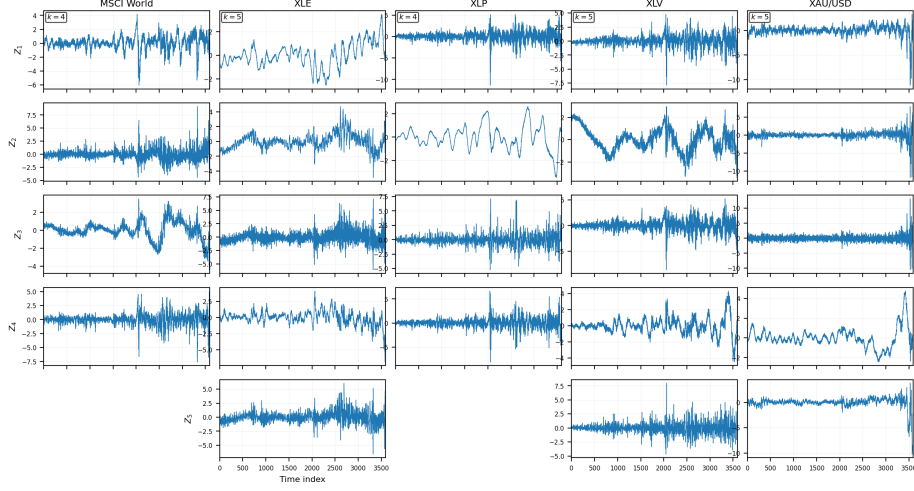


Fig. 4. FastICA scores after CEEMDAN on the empirical panel (5×5 layout). Rows index $Z_1, \dots, Z_5$; asset-specific k is annotated in each column (empty cells when $k < 5$). Columns correspond to MSCI World, XLE, XLP, XLV, and XAU/USD.

Cross-correlation among scores is numerically negligible in every case ($\max |r| < 10^{-13}$), confirming practical independence after compression. Mean IMF reconstruction R^2 ranges from 0.62 (XLP) to 0.74 (XLV), with relative Frobenius errors between 0.25 (XLE) and 0.75 (XLP). The independent scores fluctuate around zero and retain multi-scale oscillatory structure. Along with the drift component held outside the ICA fit, the decorrelated Z_j channels provide the reduced oscillatory input passed to the parallel graph encoders in Section 4.5.

4.5 Series to Graph Transformations

The graph-encoding stage converts the decorrelated ICA panel into the relational input prepared for downstream GNN modeling. Each asset is encoded separately on the aligned empirical calendar. For every retained FastICA score Z_j , the univariate channel is mapped in parallel to horizontal visibility (HVG), natural visibility (NVG), and recurrence graphs as defined in Section 3.4. Visibility rules require no free parameters beyond the input series. Recurrence delays τ , embedding dimensions d , and thresholds ε_{rec} are estimated independently for each channel. The CEEMDAN slow trend component is kept outside this operational panel: Section 4.3 shows that it carries long-horizon drift with high linear fit (R_{lin}^2 up to 0.99 on the equity benchmarks), so its evolution can be addressed with simple regression rather than graph encoders that would generate ultra-dense visibility structures.

ICA also reduces the number of graph instances relative to a native IMF baseline that applies the three encodings to each of the p oscillatory IMFs per asset

($3p$ graphs, residue excluded). Retaining k ICA scores lowers the operational count to $3k$ —that is, k channels times three encodings per Z_j (Section 4.4). Aggregated over the five eligible series, G_{IMF} falls from 123 to $G_{\text{ICA}} = 69$ ($\rho_G = G_{\text{ICA}}/G_{\text{IMF}} \approx 0.56$, a $1.8\times$ reduction in graph count). Total node mass scales in proportion ($V_{\text{tot,IMF}} \approx 4.40 \times 10^5$ versus $V_{\text{tot,ICA}} \approx 2.47 \times 10^5$, $\rho_V \approx 0.56$) because every instance has $|V| \approx T$. Aggregate link load contracts more sharply: $E_{\text{tot,IMF}} \approx 1.38 \times 10^7$ versus $E_{\text{tot,ICA}} \approx 9.7 \times 10^5$ ($\rho_E \approx 0.07$), reflecting NVG sparsification on the slow oscillatory IMFs that dominate edge growth in the native panel.

Connectivity and distance The same audit metrics used for pre-compression IMF graphs are applied to every Z_j instance in the ICA panel. For each channel we record $|V|$, $|E|$, edge density, the number of connected components, mean degree, and diameter when the graph is connected. Node sets remain one-to-one with the sample length on visibility encodings ($|V| \approx T \in \{3584, 3587, 3596\}$). Recurrence graphs shorten slightly because of delay embedding ($|V| \approx T - 30$ on average). Table 4 summarizes mean connectivity statistics averaged within each asset and encoding family over the k oscillatory ICA channels.

Horizontal visibility graphs on ICA scores form a homogeneous connectivity class across the panel. Mean $|E|$ lies between 7.0×10^3 and 7.2×10^3 per asset, with densities near 1.1×10^{-3} and mean degree ≈ 4 . Every HVG instance is connected, with diameters between 22 and 273 across channels (asset means between 40 and 96). These ranges closely match the native IMF HVG panel (mean $|E| \approx 6.9 \times 10^3$), indicating that ICA preserves the sparse convex visibility footprint of the oscillatory subspace after linear mixing.

Under natural visibility, the ICA scores remain connected on every channel but are markedly sparser than on individual IMFs. Mean NVG edge counts range from 1.4×10^4 (XLV) to 2.8×10^4 (XLP), with densities in $[1.7 \times 10^{-3}, 4.3 \times 10^{-3}]$ and diameters between 10 and 16. The IMF NVG baseline, in contrast, exhibits strong scale heterogeneity: mean $|E|$ per graph rises to $\approx 3.1 \times 10^5$ because slow IMFs generate up to 10^6 links. ICA therefore reduces aggregate NVG link load by more than one order of magnitude relative to the oscillatory IMF panel while keeping unit connectivity.

Thresholded recurrence on Z_j carries $\approx 1.6 \times 10^4$ – 1.7×10^4 edges and densities near 2.5×10^{-3} – 2.8×10^{-3} , comparable to the IMF recurrence panel ($\approx 1.7 \times 10^4$ links on average). Unlike visibility encodings, recurrence networks on ICA scores are highly fragmented: the number of connected components ranges from a few hundred to $\approx 1.6 \times 10^3$ per channel, so diameters are undefined and global distance summaries are not informative. Mean degree stays near 9–10, consistent with the thresholded ε -neighborhood construction in phase space.

Local clustering structure Local neighborhood structure is summarized by the mean, median, and standard deviation of the node-wise clustering coefficient $\bar{C}$, computed on each Z_j graph when tractable. Table 5 reports asset-level means of $\bar{C}$ averaged over the k oscillatory channels.

Table 4. Mean connectivity statistics on oscillatory ICA channels $(Z_1, \dots, Z_k)$ by asset and encoding. $|E|$ and recurrence component counts are averaged over k channels. Diameters apply to visibility graphs only.

Asset	k	HVG $ E $	NVG $ E $	REC $ E $	$\bar{D}_{\text{HVG}}$	$\bar{N}_{\text{cc,REC}}$
MSCI World	4	7161	14362	16918	40	1428
XLE	5	7110	20746	17256	73	897
XLP	4	7089	27845	16207	96	1140
XLV	5	7141	13917	16630	44	1408
XAU/USD	5	7124	15326	17338	49	798

On horizontal visibility graphs, ICA scores yield a tight clustering band across the panel ($\bar{C}_{\text{HVG}} \approx 0.59\text{--}0.66$ per channel, asset means in Table 5). Cross-channel dispersion within each asset is small (standard deviation below 0.02 on four of the five instruments), indicating that linear compression homogenizes local convex structure among mixed oscillatory coordinates. The native IMF HVG panel, by contrast, exhibits a pronounced scale gradient: $\bar{C}_{\text{HVG}}$ falls from ≈ 0.66 on fast IMFs to ≈ 0.33 on slow modes (panel mean ≈ 0.52). ICA therefore raises and stabilizes HVG clustering relative to the multi-scale IMF baseline.

Under NVG, the Z_j channels display the highest local cohesion in the ICA panel ($\bar{C}_{\text{NVG}} \approx 0.70\text{--}0.76$ per asset, panel mean ≈ 0.73). Median node coefficients cluster near 0.76–0.80, with cross-channel standard deviations typically below 0.04 except on XLP (≈ 0.08), where one mixed channel reaches $\bar{C}_{\text{NVG}} \approx 0.57$. These levels are comparable to the fast-to-medium IMF NVG range and well above the values attainable on slow native modes, whose ultra-dense NVG instances either inflate local closure or become too costly to evaluate.

Recurrence encodings exhibit lower, more heterogeneous clustering ($\bar{C}_{\text{REC}} \approx 0.21\text{--}0.34$ per asset; panel mean ≈ 0.29). Channel-wise values span 0.21–0.59, reflecting fragmented ε -neighborhoods and occasional locally cohesive patches in phase space. Median node coefficients are often near zero because most vertices lie in sparse components, while the standard deviation of node-wise clustering remains broad ($\approx 0.26\text{--}0.31$). Mean recurrence clustering on the ICA panel sits slightly below the IMF baseline average (≈ 0.34), consistent with the higher component count on mixed Z_j coordinates reported in Table 4.

Overall, the clustering audit shows that ICA compresses cross-scale heterogeneity into a narrow band of local statistics on visibility encodings, while recurrence graphs retain channel-specific phase-space texture that visibility summaries alone do not capture.

Interpretation This encoding stage yields a relational panel that is simultaneously smaller and better conditioned for downstream GNN use than an IMF encoder applied directly to the same oscillatory subspace. Inventory and node mass contract by a factor of ≈ 1.8 when p IMF channels are replaced by k decorrelated scores, while aggregate link load falls by more than one order of magnitude. The

Table 5. Mean local clustering coefficients on oscillatory ICA channels ($Z_1, \dots, Z_k$) by asset and encoding. $\bar{C}$ is averaged over k channels within each asset.

Asset	k	$\bar{C}_{\text{HVG}}$	$\bar{C}_{\text{NVG}}$	$\bar{C}_{\text{REC}}$
MSCI World	4	0.63	0.75	0.23
XLE	5	0.62	0.72	0.34
XLP	4	0.62	0.70	0.32
XLV	5	0.63	0.75	0.23
XAU/USD	5	0.64	0.74	0.32

asymmetry between these reductions is informative. ICA does not materially alter the sparse HVG footprint of the oscillatory subspace—connectivity, diameter, and $\bar{C}_{\text{HVG}}$ on Z_j track the native IMF benchmarks—so the convex visibility geometry of the compressed panel remains recognizable after linear mixing. The dominant structural gain is NVG sparsification: by centering mixed oscillatory amplitudes near zero, ICA suppresses the extreme edge growth on slow native IMFs that would otherwise dominate memory and centrality estimates, while preserving unit connectivity and locally cohesive neighborhoods on every Z_j channel.

Recurrence encodings occupy an intermediate position in this picture. Per-graph link counts remain at the same order as on the IMF panel, but fragmentation into hundreds to thousands of components shows that thresholded phase-space neighborhoods on decorrelated coordinates no longer organize into a single navigable backbone. Recurrence graphs instead complement visibility encodings rather than substituting for them: they expose dynamical recurrences and local phase-space texture that HVG and NVG summaries do not capture, at the cost of weaker global connectivity. The clustering audit reinforces this division. Visibility statistics on Z_j are tightly banded across assets and channels, whereas $\bar{C}_{\text{REC}}$ varies widely and often concentrates mass near zero because most vertices sit in small components.

Two design choices outside the graph panel itself shape the resulting architecture. First, the CEEMDAN slow trend component is modeled separately by simple regression, avoiding ultra-dense NVG constructions that add little relational information beyond monotone drift. Second, the decorrelated ICA scores validated in Section 4.4 supply statistically independent oscillatory carriers with negligible cross-correlation, so parallel graph instances are not redundant copies of the same linear dependence structure. Together, the audited ICA graph panel— $3k$ sparse-to-moderate relational objects per asset across three complementary encodings—constitutes the structured graph input whose downstream GNN use is outlined in Section 5.

5 Conclusions and Future Work

This work addressed a gap identified in Section 2.4: decomposition backend selection, oscillatory-panel compression, and scale-resolved graph encoding had evolved largely in isolation, with validation metrics centered on reconstruction or forecast error rather than on the structural properties required for downstream relational modeling. We designed and audited an end-to-end pipeline that prepares individual financial instruments for graph-based analytics under explicit budgets. Each series passes an econometric eligibility gate, is decomposed after a two-stage backend selection, is compressed with FastICA, and is encoded into a panel of visibility and recurrence graphs whose connectivity and local structure are quantified before any predictive model is trained.

What was achieved. Four results stand out from the experimental program in Section 4. First, the synthetic pathology benchmark (Section 4.2) subjected EMD, EEMD, CEEMDAN, and VMD to an identical catalog of controlled signal families and showed that no single backend dominates every regime. Coupled with residual-aware validation on a diversified panel—MSCI World, three U.S. sector ETFs (XLE, XLP, XLV), and spot gold (XAU/USD)—this evidence justified retaining CEEMDAN as the operational backend because its explicit slow trend component concentrates long-horizon drift with markedly lower roughness than EEMD or VMD ($\mathcal{R}_{\Delta^2}$ between 10^{-7} and 10^{-6} on the equity benchmarks), even when all methods close the signal numerically. Decomposition quality must therefore be judged through trend-component interpretability, not closure error alone.

Second, FastICA reduced the oscillatory IMF panel to $k \in \{4, 5\}$ statistically independent scores per asset with negligible mutual correlation ($\max |r| < 10^{-13}$) and mean IMF reconstruction R^2 between 0.62 and 0.74 (Section 4.4). This compression lowers the number of parallel encoding branches while preserving multiscale oscillatory structure in decorrelated coordinates.

Third, the graph-encoding audit (Section 4.5) quantified the structural gain relative to a native IMF baseline ($3p$ graphs, residue excluded). Aggregated over five eligible series, graph inventory fell from 123 to 69 instances ($\rho_G \approx 0.56$), node mass scaled in proportion ($\rho_V \approx 0.56$), and aggregate link load contracted far more sharply ($\rho_E \approx 0.07$, a $\approx 14\times$ reduction). ICA preserved the sparse horizontal-visibility footprint of the oscillatory subspace while suppressing the ultra-dense natural-visibility growth on slow native modes that would otherwise dominate memory and centrality estimates. Recurrence encodings retained comparable per-graph link counts but exhibited heavy fragmentation, confirming that visibility and recurrence graphs capture complementary relational views.

Fourth, two design choices keep the output panel well conditioned for downstream use. The CEEMDAN slow trend component is modeled outside the graph panel by simple regression, avoiding encoders that add little information beyond monotone drift. The decorrelated ICA scores ensure that parallel graph instances are not redundant copies of the same linear dependence structure. Together, these stages yield, per asset, $3k$ audited relational objects across HVG, NVG,

and recurrence encodings—the structured input prepared for graph neural network (GNN) modeling surveyed in the financial forecasting literature [51,18,5].

Future directions. The present study stops at graph construction and structural auditing. The natural next step is to apply predictive and explanatory analytics on the generated graph panel using GNN architectures that combine temporal encoders with graph convolutions or attention over relational structure [28,48,20,18,5,19]. Candidate tasks include multivariate return forecasting, regime classification, and volatility-state detection on the Z_j -level graph families, where HVG and NVG supply globally connected visibility summaries and recurrence graphs expose local phase-space texture. Spectral-temporal designs such as STGNN [5] and dynamic graph models [19] are particularly aligned with the multiscale, multi-encoding panel produced here.

Scalability is a second priority. ICA drives cross-correlation among oscillatory channels toward zero, so graph construction and per-channel GNN training decompose into embarrassingly parallel workloads across assets and Z_j branches—a pattern consistent with the distributed processing rationale developed in Section 2.2. As instrument coverage and sample length grow, the map-and-reduce programming model [11] provides a well-established foundation for distributing independent decomposition, encoding, and feature-extraction jobs across commodity clusters, while vertex-centric bulk-synchronous frameworks such as Pregel [37] address phases that require coordinated message passing over large graph aggregates. For deep learning at scale, mini-batch and partitioning strategies developed for large GNN training—including GraphSAGE-style inductive sampling [20] and Cluster-GCN partitioning [8]—offer practical routes to keep memory and epoch time manageable when thousands of visibility and recurrence graphs are processed jointly.

Further work should extend the empirical panel beyond five instruments, evaluate cross-asset relational graphs built from synchronized ICA panels, and benchmark end-to-end latency and forecast quality against CEEMDAN–LSTM baselines that operate on sequential IMF channels without graph encoding [6,33]. Closing that loop will test whether the structural budgets enforced upstream translate into measurable gains in predictive performance and computational efficiency when GNN analytics are deployed on distributed infrastructure.

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